PALEOMAGNETIC STRATIGRAPHY OF THE MEADOWCLIFFE TILL, SCARBOROUGH BLUFFS, ONTARIO: A LATE PLEISTOCENE EXCURSION ?

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Abstract. The 12 m thick Meadowcliffe Till was sampled in 5 vertical profiles spread horizontally over 400 m in the Scarborough Bluffs near Toronto, Ontario. This paleomagnetic study of ≃700 specimens yielded a normal segment in all 5 profiles and an excursive segment in 3 profiles. The normal zones give, in most cases, significantly different directions which suggests that they are not strictly coeval and that deposition rates in this till are highly variable from one point to the next. Overall, they give a virtual geomagnetic pole located at 273.1°W, 73.0°N (δp = 1.7°, δm = 2.6°). Similarly the 3 excursive zones are not exactly coeval. Assuming the remanence reflects a single excursion then the pole appears to swing from mid-Africa to mid-South Pacific, and then back to the normal Arctic pole. This polar sequence and the apparent paleofield intensities are notably similar to those found in aboriginal fireplaces at Lake Mungo, Australia. The Meadowcliffe Till is the first Ontario till to yield evidence of recording a geomagnetic field excursion(s). Accordingly its paleomagnetic age is set at 30,500 ± 1,500 yr B.P. which is within the 8,000 yr window permitted by the single radiocarbon dates from below and above the till.

Introduction

About 130 m of Pleistocene deposits are exposed in the Scarborough Bluffs, east of Toronto, on the north side of Lake Ontario. Their age span is from ≃70,000 to ≃11,000 yr B.P. [Dreimanis, 1977]. The Meadowcliffe Till is an interbed in the interstadial Thorncliffe sands and clays. Radiocarbon dates on plant remains in the Thorncliffe show that the till was deposited between ≃38,000 yr B.P. and 31,500 ± 1,500 yr B.P. [Dreimanis, 1977]. A geomagnetic field excursion(s) is recorded in Australian Lake Mungo aboriginal fireplaces, dated by radiocarbon at 29,500 ± 1,400 yr B.P. [Barbetti and McElhinny, 1976], in the French Laschamp lavas dated by thermoluminescence, radiocarbon, and K/A at 33,000 ± 4,000 yr B.P. [Valladas et al. 1977; Labeyrie and Gillot, 1977], and in Gulf of Mexico cores dated by radiocarbon at 32,500 ± 1,500 yr B.P. [Freed and Healy, 1974]. The Meadowcliffe Till was studied paleomagnetically to see if it recorded this excursion(s) and, if so, thereby refine the dating on the till and provide a valuable Pleistocene marker horizon.

Sampling

The 12 m thick Meadowcliffe Till is clay rich containing ≃54% clay, ≃35% silt and ≃11% sand [Karrow, 1967]. It forms nearly vertical cliffs. A dry surface crust of ≃30 cm must be removed befor sampling. At the Sunnypoint section [Karrow, 1967; A1011], 30 blocks of till measuring 10 x 10 x 15 cm each were cut out with a chisel to form a profile of the bottom 8 m of till (Fig. 1). The blocks were oriented with a level and Brunton compass. A drill press with a 2.5 cm diameter Felker thin-wall bit was used to drill 5 cores from each block. Two specimens were cut from each core and sealed with polyurethane to prevent desiccation. About 300 m to the east at the Hi section [Karrow, 1967; A1020], 4 vertical profiles totalling 200 cores were collected using a method similar to that adopted by Johnson et al. [1948] for varves. Polycarbonate plastic tubes with a 2.5 cm outer diameter and a 1.6 mm thick wall were driven ≃7.5 cm into the section face using a brass driving head to distribute the hammer impacts. The till-filled tubes were marked, labelled and pulled out. The resulting holes were oriented using a Brunton or sun compass mounted on an orienting tube. Magnetic deviations up to 10° were observed on the Brunton close to the Thorncliffe sand. Two 2.4 cm long plastic-encased specimens were cut from each core and dipped in polyurethane to seal the ends. In a few places where the till is fairly moist only one 2.4 cm long core was taken. Driving longer tubes results in excessive flowage of the till in the tube. The core was discarded if flow deformation exceed ± 10% in length.

Measurement

The natural remanent magnetization (NRM) of each specimen was measured on a Schonstedt SSM-1A spinner magnetometer. One specimen from every fifth block or tenth core in each profile was step demagnetized in 8 kA/m alternating field (AF) increments up to 80 kA/m using a Schonstedt GSD-1. In most specimens the viscous remanence is removed by 8 kA/m, a stable remanence direction is isolated between 8 and 48 kA/m, and above 48 kA/m increasingly significant anhysteretic components progressively scatter the step directions (Fig. 2). The stability index method [Symons and Stupavsky, 1974] was used to select the optimum AF demagnetizing field employed on adjacent specimens before remeasuring their remanence. Thus 8 to 10 (2) specimen mean directions were averaged to give each block (core) mean direction.

Results

The lower half of profile 1, most of profile 3 and all of profiles 4 and 5 are normal and appear to represent a coeval normal paleofield, N1 (Fig. 1). However, their mean directions are not statistically similar. Their cones of 95% confidence do not overlap (Fig. 3, Table 1).

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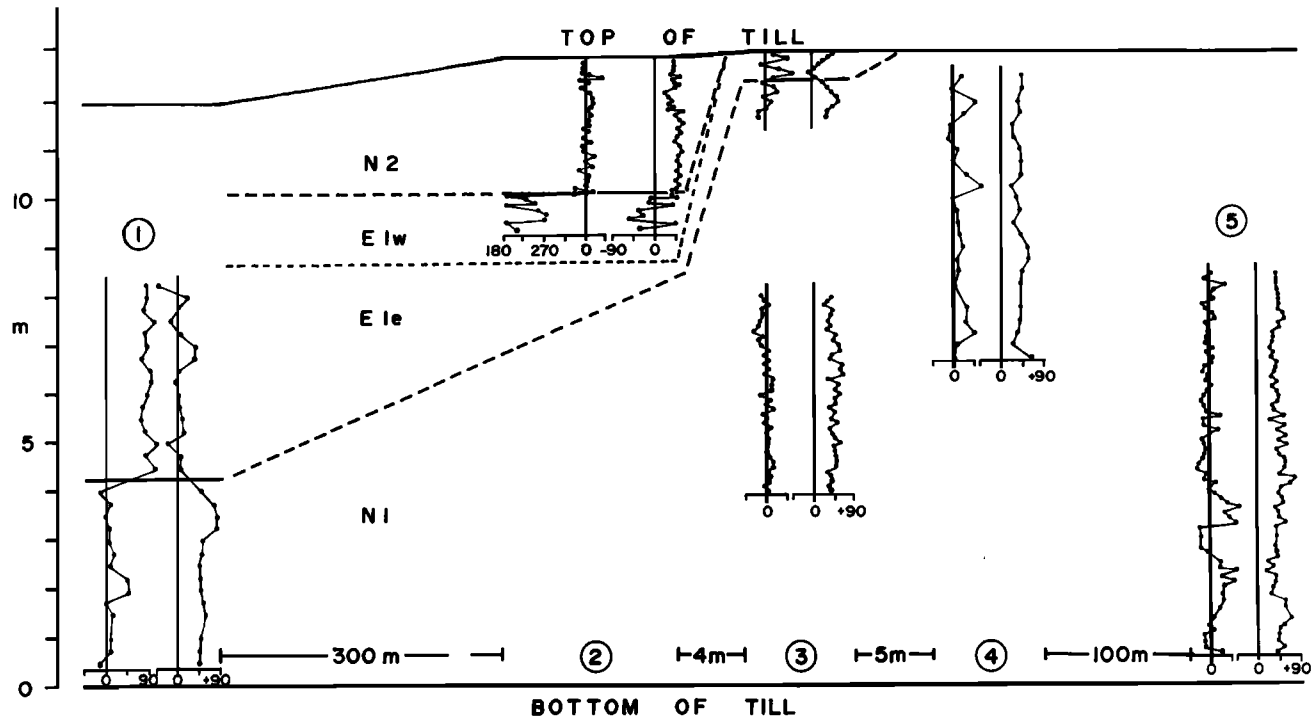


Fig. 1. Remanence profiles of the Meadowcliffe Till in the Scarborough Bluffs. Profiles 1 to 5 indicate the declination (180°-270°-0°-180°) and inclination (-90°-0°-+90°) of the remanence using block means in profile 1 and core means in profiles 2 to 5. The vertical thickness is to scale in meters and the horizontal separation is not to scale but stated in meters. The polarity zones are the basal normal (N1), the intermediate excursive with lower easterly (Ele) and upper westerly (Elw) subzones, and the upper normal (N2).

Note also that 1N1 has a stronger average intensity than 3N1, 4N1 and 5N1. In addition, there is no convincing correlation of secular variation between the N1 profiles. These factors suggest that the N1 profiles may represent slightly different times within the normal period.

The upper half of profile 1 and the top 1 m of profile 3 record a geomagnetic field excursion (Ele) giving horizontal and eastward remanence directions. The 1Ele excursion is consistent and pronounced over ≃4 m. The reality of 3Ele may be questioned. This excursion is defined in only about 8 cores in the top 1 m of the till section. Although this top 1 m of till appears undeformed in the field, it is possible that the excursion was caused by postdepositional deformation induced by overriding ice.

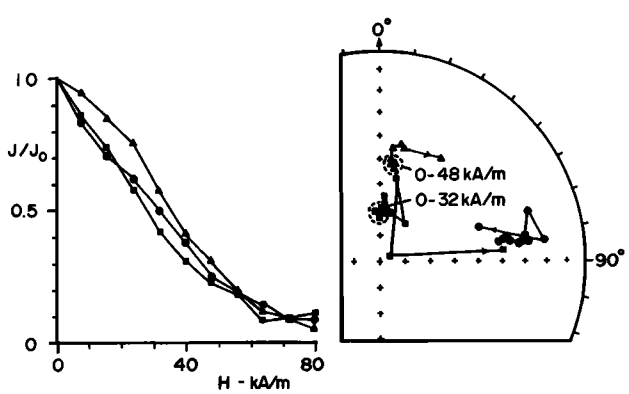


Fig. 2. AF step demagnetization results for 3 typical specimens. On the left is a remanence intensity decay plot giving the step intensity as a ratio of the initial intensity (J/Jo) versus the step peak AF demagnetizing field (H). On the right is a portion of an equal area stereonet showing the change in remanence direction with the arrow directed towards the final 80 kA/m step direction.

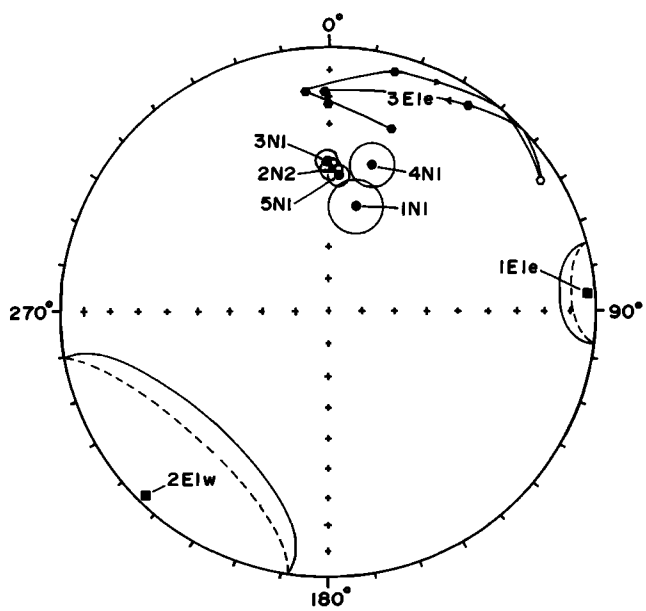


Fig. 3. Mean remanence directions for magnetic zones in the Meadowcliffe Till. Conventions as in Fig. 1, Fig. 2 and Table 1. 3Ele records the sequence of specimen mean directions.

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TABLE 1: Mean remanence directions and pole positions for the Meadowcliffe Till

Profile and Zone	Numbers of core Means	Mean Remanence Direction Declination (degrees)	Inclination (degrees)	α_{95} (degrees)	Intensity emu/cm^3 x 10^6	Pole Position Longitude ^oW	Latitude ^oN	δ_p degrees	δ_m degrees
1 N 1	14*	14.8	56.2	8.7	58.1	322.7	76.8	9.0	12.5
3 N 1	46	358.6	41.7	3.4	5.9	255.4	70.3	2.5	4.1
4 N 1	23	15.7	41.4	6.6	4.5	43.6	64.8	4.9	8.1
5 N 1	73	4.3	46.9	4.0	5.9	273.0	74.1	3.3	5.1
N 1	156	5.0	45.5	2.5	10.4	274.4	72.9	2.0	3.2
2 N 2	68	1.3	45.7	3.4	31.6	263.4	73.4	2.8	4.4
N	224	3.9	45.6	2.0	16.8	271.3	73.0	1.7	2.6
1 E 1e	16*	85.8	2.1	11.2	20.6	347.1	3.8	6.6	11.2
3 E 1e	11	19.4	22.7	14.3	8.3	292.8	54.0	8.1	15.2
2 E 1w	11	224.1	3.8	36.1	31.3	132.7	-29.7	18.1	36.1

*block means

Such local deformation could account for the failure to observe E1e in profile 2 only 5 m away. It is also possible that time lines within the till are not horizontal due to differential rates of erosion and/or deposition.

The bottom 1 m of profile 2 also records a geomagnetic field excursion (E1w) giving southwesterly and partially reversed remanence directions. Three points should be made. First, the reversed directions in this population are fairly consistent. The 2 normal cores in the 2E1w zone are single specimen cores from moist till and may provide evidence for resetting of the magnetization during sampling. If so, then the thixotropic remanent magnetization mechanism reported by Games and Wilson [1977] for adobe bricks may be the cause, with the hammer blows on the plastic tube producing instability. Second, assuming a single excursion, then the pole moves from the Arctic to equatorial Africa, to the mid-South Pacific Ocean, and back to the Arctic. The remanence intensity is lower in 1E1e, constant in 2E1w and higher in 3E1e suggesting a marked variation in the intensity of the paleofield during the excursion. The Lake Mungo excursion [Barbetti and McElhinny, 1976] also gives mid-African (F7 and F12) and mid-South Pacific (F6, F8 and F9) poles as well as decreased, constant, and increased paleofield intensities during the excursion. Third, this is the only till of the 5 that we have studied to date from Ontario [Stupavsky et al., 1974 a, b; Gravenor and Stupavsky, 1976] that gives evidence of having recorded one or more excursions of the geomagnetic field. For this reason it is a notable paleomagnetic marker horizon. Because of its striking similarity to the 29,500 ± 1,400 yr B.P. Lake Mungo excursion which overlaps the 31,500 ± 1,500 yr B.P. minimum limit for the Meadowcliffe Till [Dreimanis, 1977], we prefer to set the age of the Meadowcliffe excursion at an intervening 30,500 ± 1,500 yr B.P. The Gulf of Mexico excursion at 32,500 ± 1,500 yr B.P. could also be correlative although only remanence inclinations are available for this core. Similarly the 33,000 ± 4,000 yr Laschamp excursion could be correlative realizing that it does not have tight magnetostratigraphic control. The possibility of several excursions during a period of paleofield instability around 30,500 yr B.P. is also plausible.

The uppermost 3 m of profile 2 records the return of the paleofield, after a period of excursive instability, to a normal (N2) position centred squarely in the middle of the N1 population.

The lack of correlation of the remanence characteristics over the short interprofile separations is disturbing. The N1 zones of profiles 1 to 5 do not appear to be strictly coeval. The E1e zone in profiles 1 and 3 (?) is not observed in the intervening profile 2 covering apparently equivalent stratigraphic levels in the till. Profile 2 is the sole profile to record E1w and N2 despite being only a few meters from the stratigraphically equivalent profiles 3 and 4. Therefore, more complex correlations may well be correct rather than the simple assumed correlation of the excursive E1 zones. We also conclude the deposition of this till was non-uniform with rapid and variable rates of deposition from point to point within the till.

In concert with pole positions determined previously for southern Ontario tills [Stupavsky et al., 1974 a,b; Gravenor and Stupavsky, 1976], the pole position for the Meadowcliffe Till is far-sided and in north central Russia (Table 1). This coincidence may reflect the real paleofield directions or it may simply record climatic or sedimentolo-

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gic effects [Wollin et al., 1971; King, 1974; Bucha, 1977; and Kent and Opdyke, 1977].

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